Answer Position and Lexical Pruning Shape Synthetic Clause Retrieval in a Small Language Model

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Abstract

Does adding context hurt a small language model, and do the location and topicality of that context matter? We test the frozen Qwen2.5-0.5B-Instruct Q8 artifact on 120 deterministic synthetic contract-clause probes spanning mean prompt lengths of 1989, 3986, and 7992 tokens. Moving the same answer-bearing clause from the start to the end raised accuracy by 58.3 percentage points at the shortest tier and 50.0 points at the middle tier, while keyword pruning lowered matched accuracy by -33.3 points. The matched amount comparison was inconclusive: accuracy rose by 8.3 points and then changed by 0.0 points, with only 12 shared cases at each tier. For this model and task, end placement improved retrieval and a keyword filter amplified the lexical trap, while the amount experiment could not establish whether longer context helped or hurt.

1. Introduction

You paste a long contract into a chatbot and ask, "What code renews the vendor agreement?" The answer sits in one clause, but a similar buyer clause repeats the words in your question and gives a different code.

Long prompts make this ordinary lookup less predictable. A model may fail because the document is longer, because the answer appears in a weak position, or because nearby text resembles the question. Those explanations imply different choices: shorten the prompt, move the key clause, or filter the document. A useful evaluation must vary each property while holding the others fixed.

We study these choices in a synthetic, gradeable version of the contract situation. Each document contains an answer-bearing vendor clause and a look-alike clause for another company. The question uses the wording of the look-alike clause, while the answer clause expresses the same relation with different words. This construction reduces the value of literal matching, following the motivation of NoLiMa (Modarressi et al., 2025).

Our contribution is a fully paired, laptop-scale evaluation of a frozen open model. Prior work established position and length failures in larger systems (Liu et al., 2023; Levy et al., 2024), and showed that irrelevant statements can disrupt reasoning (Shi et al., 2023). We instead compare topically related and unrelated filler at matched prompt lengths, positions, and questions. We also test whether a deterministic keyword filter helps the exact cases on which the unfiltered model was evaluated.

The experiment asks four questions:

1. As the document grows from 1989 to 7992 mean tokens, how much does accuracy change on shared cases?
2. How does moving the same answer clause among the start, middle, and end change accuracy?
3. Does topically related filler hurt less than unrelated filler in matched documents?
4. Does keyword pruning recover accuracy on matched full-context cases?

The answer is not a general context-length threshold. In this grid, the model improved when the answer appeared near the end, related filler was no safer than unrelated filler, and keyword pruning amplified the lexical trap built into the probes.

2. Related Work

Liu et al. (2023) found a U-shaped position curve in

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multi-document question answering and key-value retrieval, with stronger use of information near context boundaries. Our design performs a smaller clause-reordering test: the question, filler sequence, clauses, and codes remain fixed while the answer clause is inserted at a different depth. Inserting it before the competing clause shifts that clause's final sentence index by one. The observed pattern is primarily recency rather than a full U-shape because the start condition is also weak.

Input amount can matter even when the underlying problem is fixed. Levy et al. (2024) padded matched reasoning problems and observed degradation well before model context limits, while RULER found that usable context lengths can be shorter than advertised windows (Hsieh et al., 2024). Our amount comparison uses the intersection of cases present at every tier, avoiding a change in scenario mix as prompts grow.

Shi et al. (2023) added irrelevant statements to arithmetic problems and found lower reasoning accuracy. We ask a different quality question: whether filler from the same business domain is less harmful than unrelated prose when amount and question wording match. NoLiMa shows why literal needle tests can overstate retrieval ability (Modarressi et al., 2025); our answer clause therefore paraphrases the question while a competing clause shares its surface form. The tested model belongs to the instruction-tuned Qwen family (Yang et al., 2024).

3. Probe Design

Model and execution. The subject is the frozen qwen2.5-0.5b-instruct-q8_0.gguf artifact, with its SHA checksum shown below:

ca59ca7f13d0e15a8cfa77bd17e65d24f6844b554a7b6c12e07a5f89ff76844e

We run it through llama-completion with greedy decoding, zero temperature, and a fixed seed. The complete manifest contains 120 probes, each stored as one prompt and one raw output. Existing outputs are skipped on rerun, so interrupted inference resumes by stable probe identifier.

Scenario families. Seeded code generates a company, target code, competing company and code, question, and filler for each family. The question asks for the renewal code needed to extend the target company's contract. Its wording exactly matches the competing clause, whereas the answer-bearing clause calls the same value a renewal identifier. A matched literal control reverses these phrasings while retaining the same filler.

For each full-context family, we insert the answer clause near the start, middle, or end. Family generation does not use target position in any random seed. The competing clause receives an earlier or later fractional insertion index that is independent of answer position and fixed across the three renders. Its final sentence index shifts by one when the answer is inserted before it. Thus a paired position contrast changes answer placement and this mechanical one-sentence shift, while preserving the scenario content and filler order.

Amount and quality. The tokenizer, rather than a character estimate, determines prompt length. The three full-context tiers average 1989, 3986, and 7992 tokens. Each tier balances topically related filler with unrelated sentences. Related filler consists of other companies, business records, and unrelated codes; unrelated filler describes places and events outside the contract domain. Quality pairs share the family, target position, question, and measured tier.

The expensive final tier contains fewer families. Amount estimates therefore use only the 12 cases shared by every tier, not the unequal full-tier totals. Quality estimates use 18, 15, and 6 pairs at the three tiers, respectively.

Pruning and controls. At the middle tier, a keyword filter retains sentences with at least two content words shared with the question, falling back to the highest-overlap sentences if too few qualify. It preserves original order. The comparison is restricted to the 30 cases with both full and pruned outputs. We also include 6 closed-book questions and 6 matched literal-versus-paraphrase controls.

4. Scoring and Comparisons

Normalized accuracy requires the target code to appear without the competing code. Exact-output compliance requires a response containing one code and no other text. A distractor capture contains the competing code without the target, and an invalid response contains both expected codes or neither. We strip only the runner's fixed transport trailer before applying these rules.

The inferential target is the seeded generator's population of synthetic scenario families. Every 95% interval resamples whole observed families, retaining all positions, filler types, and variants from each selected family; exact enumeration makes this cluster bootstrap deterministic. Paired changes are differences in accuracy over an exact intersection of scenario keys, with the same family resampling used for their intervals. These

Table 1. Matched amount and filler-quality comparisons. Brackets give 95% family-cluster bootstrap intervals; n is the matched denominator. Changes are later minus earlier for amount and related minus unrelated for quality.

Comparison	Accuracy A	Accuracy B	Change (n)
Amount: short to middle	50.0 [50.0, 50.0]	58.3 [35.2, 81.5]	8.3 (12)
Amount: middle to long	58.3 [35.2, 81.5]	58.3 [50.6, 66.0]	0.0 (12)
Quality at short tier	44.4 [22.2, 61.1]	22.2 [11.1, 33.3]	-22.2 (18)
Quality at middle tier	46.7 [6.7, 86.7]	33.3 [13.3, 53.3]	-13.3 (15)

Table 2. Accuracy by answer-clause position. Each row compares the same scenario families across positions. Brackets give 95% family-cluster bootstrap intervals.

Mean tokens	Start	Middle	End
1989	16.7 [0.0, 33.3]	8.3 [0.0, 25.0]	75.0 [58.3, 91.7]
3986	20.0 [0.0, 40.0]	30.0 [0.0, 70.0]	70.0 [30.0, 100.0]
7992	25.0 [1.9, 48.1]	75.0 [51.9, 98.1]	75.0 [51.9, 98.1]

intervals describe sensitivity to the small family sample, while wins, losses, and ties expose the underlying paired outcomes. They do not assign sampling uncertainty to greedy decoding, which is deterministic.

5. Results

Amount. The amount effect remains unresolved. Accuracy was 50.0% at 1989 mean tokens and 58.3% at 3986, a paired change of 8.3 points with a family-cluster interval of [-14.8, 31.5] over 12 cases. It was 58.3% at 7992, with 2 wins, 2 losses, and 8 ties relative to the middle tier (Table 1). Because the longest-tier intersection represents only 12 probes from 2 scenario families, these observations neither establish a benefit nor rule out degradation within the measured range.

Position. Moving the answer clause to the end produced the largest and most consistent gain. At the shortest tier, accuracy rose from 16.7% at the start to 75.0% at the end, a paired increase of 58.3 points with an interval of [50.0, 75.0], 7 wins, and no losses. At the middle tier, the same move raised accuracy by 50.0 points [20.0, 80.0], with 5 wins and no losses. The sparse longest tier also favored the end over the start by 50.0 points, but it contains only 4 pairs. Middle performance was below end performance at the two populated tiers (Table 2).

Filler quality. Topically related filler was more harmful, not safer. Relative to matched unrelated filler, related business clauses changed accuracy by -22.2 points [-33.3, 0.0] over 18 pairs at the shortest tier and by -13.3 points [-46.7, 13.3] over 15 pairs at the middle tier. The respective paired outcomes were 1 win, 5 losses, and 12 ties, then 1 win, 3 losses, and 11 ties. The longest-tier estimate was -16.7 points, but its 2 families are too sparse to strengthen that conclusion.

Table 3. Matched keyword-pruning result and diagnostic controls. Accuracy intervals are 95% family-cluster bootstrap intervals.

Condition	Accuracy	Count
Full context	40.0 [13.3, 66.7]	30
Keyword pruned	6.7 [0.0, 20.0]	30
Literal wording	83.3 [50.0, 100.0]	6
Paraphrase wording	16.7 [0.0, 50.0]	6
Closed book	0.0 [0.0, 0.0]	6

Mitigation. Keyword pruning backfired on the matched cases. Full-context accuracy was 40.0%, compared with 6.7% after filtering, a change of -33.3 points [-63.3, -3.3] across 5 families. Only 1 case improved; 11 worsened and 18 tied (Table 3). The mechanism follows from the probe design: the filter retains the competing clause because it repeats the question's words, while the paraphrased answer clause has less lexical overlap.

The controls support this explanation without proving it for other tasks. Literal wording improved matched accuracy by 66.7 points, with 4 wins and no losses. Across all outputs, distractor capture was 60.0%, exact-output compliance was 95.0%, and invalid output was 8.3%. The failures therefore arose mainly from selecting the competing code, not from ignoring the requested format.

6. Limitations

The findings concern one small quantized model and may not transfer to larger models or other quantization levels. Synthetic clause lookup is not natural contract analysis, and retrieval accuracy is not a measure of general capability. The 45-minute CPU budget capped the grid, leaving only 12 full-context probes at the longest tier and wide intervals throughout. That sparse tier prevents a conclusion about degradation within the measured range, and the experiment cannot address longer prompts. Greedy decoding removes run-to-run sampling variation but does not show how sampling would change the estimates. The mitigation is a deliberately simple keyword filter rather than a learned retrieval system.

7. Reproducibility

Every reported quantity is generated by analysis scripts from the committed probe manifest and raw model outputs, following the reporting principle that computational evidence should expose its budget and variation (Dodge et al., 2019). The model and cached data are frozen and checksummed, and each raw output retains its stable probe identifier. Running make all regenerates results.json, the LaTeX value macros, and the paper from cached outputs; running make clean-deep && make all repeats deterministic model inference. Repository gates check frozen inputs, citation resolution, number provenance, pipeline freshness, result sanity, prose style, anonymity, and compilation.

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